Bihar Engineering University, Patna
B.Tech. 7th Semester Examination, 2023

Course: B.Tech.
Code: 101722

Subject: Foundation Engineering

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory*

Q.1 ***Choose the correct answer of the following (any seven):*** **[2 x 7 = 14]**

(a) According to IS codes, the maximum differential settlements should not exceed _____ in the foundation on clayey soil.
- (i) 25 mm
- (ii) 100 mm
- (iii) 25 mm
- (iv) 40 mm

(b) Based on the function, piles can be classified into _____ types.
- (i) 4
- (ii) 6
- (iii) 8
- (iv) 3

(c) A shallow foundation is defined as a foundation which has
- (i) Low bearing capacity
- (ii) Has a depth of embedment less than its width
- (iii) Is resting on ground surface
- (iv) Causes less settlement

(d) Lime stabilization is very effective in treating
- (i) Sandy soils
- (ii) Silty soils
- (iii) Non plastic soils
- (iv) Plastic clayey soils

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(e) Bearing capacity of a soil strata supporting a footing of size 3m x 3m will not be affected by the presence of ground water table located at a depth which is
- (i) 1.0 m below the base of the footing
- (ii) 1.5 below the base of the footing
- (iii) 2.5 below the base of the footing
- (iv) 3.0 below the base of the footing

(f) Terzaghi's equation of ultimate bearing capacity for a footing may be used for square footing resting on pure clay soil with correction factor.
- (i) 0.4
- (ii) 0.6
- (iii) 1.2
- (iv) 1.3

(g) For determining the ultimate bearing capacity of soil, the recommended size of a square bearing plate to be used in load plate test should be 30 to 75 cm square with a minimum thickness of
- (i) 5 mm
- (ii) 10 mm
- (iii) 15 mm
- (iv) 20 mm
- (v) 25 mm

(h) A grillage foundation
- (i) is provided for heavily loaded isolated columns
- (ii) is treated as spread foundation –
- (iii) consists of two sets of perpendicularly placed steel beams
- (iv) all the above.

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(i) The gross pressure intensity of a structure is
(i) Total pressure at the base of the footing
(ii) Excess pressure after the construction of the structure
(iii) Minimum pressure at the base
(iv) all the above

(j) When a footing fails due to insufficient bearing capacity, distinct failure patterns are developed depending upon
(i) Failure mechanism
(ii) Plastic equilibrium
(iii) shear
(iv) all the above

Q.2 (a) What is raft foundation? Under what circumstances raft foundation is adopted? [7]
(b) Explain the factors which affect the bearing capacity of soils. [7]

Q.3 (a) Explain the different classification of piles with neat sketch. [7]
(b) Discuss briefly the method of estimating the individual and group capacity of piles? [7]

Q.4 (a) Explain Terzaghi's analysis for determining the safe bearing capacity of the soil. [7]
(b) Calculate the ultimate bearing capacity per unit area of a strip footing 1m wide and square footing of 3m x 3m. Unit weight of soil the soil is 1.8t/m^3, cohesion -2 t/m^2 and ϕ = 20 degree, N_c =17.5, N_q = 7.5 and N_γ = 5. [7]

Q.5 (a) Explain the Rankine's theory of active earth pressure for submerged backfill. [7]
(b) Calculate the active earth pressure given that c = 20kN/m^2. And unit weight of the soil being 20kN/m^3 for a vertical cut of depth 3m. The soil is cohesionless soil. [7]

Q. 6 (a) Design a reinforced concrete rectangular combined footing for two columns A and B located 3.6 m apart. The sizes of column are 400mm x 400mm and 600mm x 600mm and the loads on them are 1000 kN and 1500kN respectively. The projection of the footing beyond the axis of the column is limited to 590mm. The limiting bearing capacity of soil is 420kN/m^2. Use M15 concrete and Fe 415 steel. [7]
(b) Explain the design aspect of trapezoidal combined footing. [7]

Q.7 (a) State the design requirement of a foundation. Discuss briefly the causes of settlement. [7]
(b) Write down the procedure to conduct standard penetration test. [7]

Q.8 (a) Write Short notes on forces acting on pile cap and under reamed piles [7]
(b) Discuss briefly earth pressure at rest, Active earth pressure and passive earth pressure with expression and neat sketch. [7]

Q.9 Explain the concept of ground movement and its causes in the context of construction activities such as excavation, tunnelling and foundation installation. [14]

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